ISTEC TRAINING · INCIDENT INVESTIGATION TOOLKIT

Quick guide

Three offline investigation tools on one USB stick: Storyline Casebook, Tripod Beta Investigator and RCA Fault Tree, with a START HERE dashboard and case register. Version 1.0 (2026-10-23).

1. Starting

1. Copy the whole `Incident Investigation Toolkit` folder to the USB stick and keep the folder structure.
2. Open `START HERE.html`, or double-click `Start Toolkit – Windows.cmd` (opens Microsoft Edge) or `Start Toolkit – Mac.command` (opens Chrome or Edge). On a Mac, the first time, right-click the file and choose Open.
3. Choose a tool. Each tool also opens directly from the `Tools` folder.

Use **Google Chrome** or **Microsoft Edge**. They can keep a case file on the stick linked to the tool, so every change is saved to it. Firefox and Safari run the tools, but there you download a copy and open it again later.

2. Saving on the stick

Tool	Save to the stick	File type
Storyline Casebook	Workspace bar at the bottom: **Save to file…**. One workspace file can hold several cases.	`.casebook-ws.json`
Tripod Beta Investigator	Header button **Save to file…** while a case is open.	`.tripod.json`
RCA Fault Tree	**File ▸ Save as…**	`.rcaft.json`

- Save into the `My cases` folder. After that, the tool writes every change to the file and the header shows **Saved to …**.
- Each tool also keeps a copy in the browser of the computer you are using. That copy stays on that computer.
- On another computer, open the file from `My cases` (in the tool, or from the case register). If the tool shows **Reconnect file**, click it: after a restart the browser asks for permission again.
- Take a spare copy now and then (Download backup), and eject the stick safely.

3. The dashboard and case register

- **Choose the My cases folder** once; the dashboard remembers it on that computer. It lists every case file in the folder, plus cases that exist only in this computer's browser (marked **This computer only**: save them to the stick).
- Summary tiles show open investigations, open and overdue actions (due date before today), and how many cases are not yet on the stick.
- **Open** passes the file to the right tool. The tool asks you to confirm with one click, because the browser needs permission to write to the file.

4. Which tool when

Tool	Use it to	Key outputs
Storyline Casebook	Run the investigation file: initial report, evidence register with photos, ECFA+ storyline of events and conditions, ICAM coding, root causes, corrective actions.	Case report, draw.io storyline
Tripod Beta Investigator	Model the incident as trios (Agent, Object, Event), place failed, missing and effective barriers, and trace each failure to Immediate Cause, Precondition and Underlying Cause (BRF).	Evidence plan (Excel checklist, photography schedule, assumptions to test), draft tree, Tripod tree, BRF profile, SMART actions, Word report, Casebook package
RCA Fault Tree	Lay out every way the top event could occur, confirm or rule out each branch with evidence, and use optional probabilities to rank the open hypotheses.	Fault tree, credible causal combinations, P(cause | incident), Word report, Casebook package

Tripod Beta and the RCA Fault Tree both export a **Storyline Casebook package**. Import it in the Casebook (Import Case Package on the case list), so root causes and actions end up in one case file.

5. Storyline Casebook

- Cases follow eight stages: initial report, evidence, description, storyline, ICAM, root causes, corrective actions, report.
- The built-in sample case shows a complete investigation. The example workspace in `Examples` holds the Tripod Annex 8 vehicle roll-over imported as a Casebook case.
- Opening a workspace file replaces the cases in this browser with those in the file. Save the current workspace first if it holds work that is not in a file.

6. Tripod Beta Investigator

- Start on the **Evidence plan** tab: write everything you know or believe about the incident, including assumptions ("I assume…") and early conclusions. **Build starter plan** turns it into an evidence checklist (Position, People, Parts, Paper, Electronic data; perishable items marked Urgent), a photography schedule in shooting order and a list of statements to test. **Export to Excel** gives a printable workbook for the field.
- For a plan or a first tree written by Claude, use **Copy prompt for Claude** (or **Copy tree prompt**), paste it into a Claude chat, then bring the reply back with **Paste Claude's answer** (or **Paste Claude's tree**). Every drafted tree element is marked assumed until the evidence confirms it.
- Drag elements from the palette onto the canvas. To link two elements, drag the round handle on one onto the other; links the tree rules forbid are refused with the reason.
- Select a line to insert a barrier on it. The inspector has quick-add buttons for the next element (Immediate Cause → Precondition → Underlying Cause).
- The **Rules check** tab lists errors, warnings and tips against the Tripod Beta tree rules (User Guide Annex 2 and 6). Each Failed or Missing Barrier and each Underlying Cause needs an action.
- Two reference cases open read only: the Annex 8 vehicle roll-over (complete) and the **TB-EX-09 dam drowning training case** (core diagram and evidence plan only). **Make a practice copy** to edit either; in TB-EX-09 you add the barriers, causation paths and actions yourself.

7. RCA Fault Tree

- The top event is the incident. Add gates (OR: any input causes it; AND: all inputs needed; INHIBIT: input plus enabling condition; VOTE: k of n) and events.
- Every element is a hypothesis: **Confirmed**, **Likely**, **Unverified** or **Ruled out**, with evidence and a planned test.
- Probabilities are optional. Each event picks its own source: safeguard/IPL PFD, human error probability, equipment failure on demand, initiating event frequency, manual value or site data. A looked-up value can be edited for one event (record why), or overridden for the whole case on the Data library tab.
- **Results** ranks the open hypotheses by P(contributed | incident) and lists the credible causal combinations (minimal cut sets) left after the ruled-out causes are removed.
- See `Guides/RCA Fault Tree – User guide.pdf` for the method and formulas.

8. Troubleshooting

What you see	What to do
“Not saved to a file” / “Save to file…”	The case is only in this browser. Save it into `My cases`.

“Reconnect file”	Click it and allow access. The browser forgets permissions when it restarts.
A case is missing on another computer	It was only in the first computer’s browser. Open that computer again and save the case to the stick.
The case register is empty	Click **Choose the My cases folder** and select the folder on the stick. In Chrome or Edge, click **Refresh** after saving new files.
Double-clicking the .command file does nothing (Mac)	Right-click it, choose Open, and confirm.

The screening values in the RCA Fault Tree library are generic, order-of-magnitude values for ranking hypotheses, not design data. The Tripod Beta method is summarised from the Tripod Beta User Guide (Stichting Tripod Foundation).